

ploidyspec

Reading polyploidy, its origin and its return to diploidy
from haplotype-resolved genome assemblies

A guide to the method, the tool, its outputs,
and what the first 45 assemblies show

What this document is

A single place to understand the whole project: why it exists, how the software works, what every output means, how to use the outputs to answer concrete questions about a genome, and what we have found so far, including what we got wrong and corrected.

Status

Written 7-8 October 2026. The software is on `main`. The panel was re-run after the fixes described in Chapter 8, so the results chapter reflects the current method.

Contents

1	At a glance	3
2	Background: why this question, why now	4
2.1	Polyploidy is common, and its categories are blurry	4
2.2	Rediploidization: the time axis of the continuum	4
2.3	Homoeologous exchange blurs things from the other side	4
2.4	Divergence records isolation, not duplication	4
2.5	Other ways to tell auto from allo, and how ploidyspec differs	5
3	What we are trying to achieve	6
3.1	The aim	6
3.2	Design principles	6
4	The tool	7
4.1	Input	7
4.2	Installing and running	7
4.3	The measurement underneath everything	7
5	The modules, one by one	9
5.1	prepare	9
5.2	kmers	9
5.3	matrix: copy distances and the relabelling check	9
5.4	homeologs: are some chromosome numbers duplicates of each other?	9
5.5	windowed: what happens along each chromosome	9
5.6	structure: genome-wide partitions and pair synchrony	10
5.7	te-markers (optional): fossil repeat activity	10
5.8	rediploidization: fusions and per-chromosome states	10
5.9	report, panel, simulate, cleanup	11
6	Answering the four questions	12
6.1	Question 1: what is the ploidy?	12
6.2	Question 2: auto-like or allo-like, and with what confidence?	12
6.3	Question 3: TE markers, explained	13
6.4	Question 4: is there evidence of rediploidization?	14
7	What we have found so far	16
7.1	Validation on simulated genomes	16
7.2	The snow carps: blind recovery of a published result	16
7.3	The panel: 45 assemblies	17
8	Caveats, corrections and open questions	19
8.1	What we got wrong, and fixed	19
8.2	Limits of the approach	19
8.3	Where ploidyspec sits in poly-space	19
8.4	Open questions	20
9	Output reference	21
10	Glossary	22
11	References	23

1 At a glance

Give ploidyspec the haplotype-resolved assembly of one individual (two or more haplotype FASTA files, or one file with haplotypes named in the headers) and it answers four questions, each with the evidence behind it:

Question	How ploidyspec answers it	Where to look
1. What is the ploidy?	Counts the haplotype copies of each chromosome, then looks for chromosome numbers that are themselves duplicated (homeolog pairs). Copies × duplicated sets gives the ploidy relative to the base number.	<code>matrix/</code> , <code>ploidy_summary.tsv</code> , <code>homeologs/</code> , <code>homeolog_pairs.tsv</code> , <code>panel_summary.tsv</code>
2. Does it behave like an auto- or an allopolyploid, and how sure are we?	Measures whether the copies of each chromosome are interchangeable (tetrasomic-like) or fall into two lineages that do not mix (disomic-like), and whether that pattern is consistent across chromosomes. Several independent lines of evidence give the confidence.	<code>rediploidization/</code> , <code>structure/</code> , Section 6.2
3. What do the TE markers say?	Counts repeat (transposable element) k-mers that are abundant in one copy but not another: a fossil record of repeat activity in separate lineages. Read as a contrast within a genome, not as an absolute score.	<code>subgenomes/</code> , Section 6.3
4. Is there evidence of rediploidization?	Looks for chromosome fusions between copies, chromosomes whose copies have split into lineages along all or part of their length, and ancient pairs that resolved at different times.	<code>rediploidization/</code> , Section 6.4

It does all of this from DNA sequence alone: no gene annotation, no reference genome, no read data, no alignment. The core measurement is the k-mer distance between chromosome copies, whole and in windows.

The one thing to remember

Sequence from a single individual shows how its chromosome copies relate to each other today. It shows inheritance-like structure, not inheritance itself (that needs crosses or populations), and not origin. A long rediploidized autopolyploid can look exactly like an allopolyploid. The outputs are worded accordingly: “tetrasomic-like”, “resolved lineages”, never “is an autopolyploid”.

2 Background: why this question, why now

2.1 Polyploidy is common, and its categories are blurry

Whole-genome duplication (WGD) has happened again and again in the history of plants and animals, and nearly all flowering plants descend from polyploid ancestors. Polyploids are traditionally sorted into two kinds. **Autopolyploids** double a genome within one species; their chromosome copies are near-identical and pair freely at meiosis (**polysomic**, or **tetrasomic** in a tetraploid, inheritance). **Allopolyploids** combine the genomes of two diverged species; each chromosome has a preferred partner from its own parent (**disomic** inheritance), and the two parental sets remain as **subgenomes**.

Twyford et al. (2025) argue that this binary is better seen as a **continuum**, or a multidimensional “poly-space”, because the definitions (by origin, by meiotic pairing, by inheritance) often disagree, and because genomes move through that space over time. Their Box 2 makes the point that matters most here: after duplication a genome undergoes **diploidization** (or **rediploidization**), and a diploidized autopolyploid can have diverged duplicate chromosomes “indistinguishable from those in a classic allopolyploid”. They suggest measurable axes to place genomes in poly-space: heterozygosity, sequence divergence between duplicated sets, gene-level divergence, and structural variation, ideally from haplotype-resolved genomes. ploidyspec measures several of these directly (Section 8.3).

2.2 Rediploidization: the time axis of the continuum

Xie et al. (2026) caught rediploidization near its start. Snow carps (Schizothoracinae) share one autotetraploid origin. In *Schizopygopsis younghusbandi* most of the 25 chromosome quartets are still tetrasomic: their four copies are about equally similar ($K_s \approx 0.01$). But on ten chromosomes, two of the four copies have fused with another chromosome (five fusions). On those chromosomes, fused copies pair with fused copies and unfused with unfused, so the quartet has split into two lineages that diverge. Divergence is highest at the fusion site and spreads outwards along the arms, and the fusions happened in three “waves” of different ages. A separate mechanism, a centromeric inversion, made the short arm of chromosome 17 disomic while the long arm stayed tetrasomic. The authors also note that downstream consequences of rediploidization, such as biased gene loss and expression between the two lineages, mimic subgenome dominance in allopolyploids.

This is the pattern ploidyspec was built to read: per chromosome, and along each chromosome, whether the copies still behave as one pool or have separated. The snow carp genomes are part of our panel, and we recover their fusions and their wave order without being told where to look (Section 7.2).

2.3 Homoeologous exchange blurs things from the other side

Allopolyploids do not stay neatly separated either. Deb et al. (2023) review homoeologous exchange (HE): recombination between chromosomes of different subgenomes. HE can be reciprocal, or “with replacement”, where one subgenome’s segment is overwritten by a copy of the other’s, homogenizing that region. HE is common in young allopolyploids, concentrated towards chromosome ends, often biased towards one subgenome, and leaves a mosaic. For ploidyspec this means an allopolyploid can show local blocks where homeologs are near-identical. Those blocks look like the regional splits of a rediploidizing autopolyploid seen in a mirror.

2.4 Divergence records isolation, not duplication

A preprint by Gaynor et al. (2026, bioRxiv; not yet peer reviewed) makes an argument that applies to any divergence-based method, including ours. Using haplotype-phased polyploid assemblies, they find that synonymous divergence (K_s) builds up only between gene copies that do not recombine. Copies on chromosomes that pair at meiosis stay similar, whatever the ploidy. Duplicate-gene divergence therefore detects meiotic isolation, or divergence inherited from hybridizing parents, rather than genome doubling as such. That is why tetrasomic autopolyploid copies stay close.

ploidyspec's distances behave the same way, and we use this deliberately. Low distance among all copies of a chromosome means the copies still mix. Two clusters at a consistent depth mean two lineages have been isolated from each other. Divergence alone cannot say whether the isolation came from hybridization (allo) or evolved later (rediploidized auto). It also explains a blind spot: an old autopolyploidy that stayed tetrasomic leaves little divergence to find.

2.5 Other ways to tell auto from allo, and how ploidyspec differs

- **Gene-loss balance (P-index; Wang et al., 2019).** After duplication, an allopolyploid tends to lose genes unequally between subgenomes (biased fractionation), and an autopolyploid equally. Wang et al. turn this into a 0-1 index from gene collinearity against an outgroup genome, with about 0.3 separating auto from allo. It needs gene annotation, synteny, and a suitable outgroup. Xie et al. found biased gene loss towards the fused copies in an autopolyploid, so biased fractionation can also arise after rediploidization.
- **Duplicated BUSCOs and gene clusters.** McHale et al. (2025) detect the ancient land snail and slug WGD from the proportion of duplicated BUSCO genes (16-21% in Stylommatophora versus 0.5-1.5% in other molluscs). They follow it in the patchwork retention of duplicated Hox clusters, mirroring vertebrates. These approaches suit old duplications whose copies have diverged beyond the reach of k-mer comparison.
- **k-mer spectra from reads** (for example GenomeScope 2.0 and Smudgeplot, Ranallo-Benavidez et al., 2020; Tetmer, Becher et al., 2020) estimate ploidy and subgenome divergence without an assembly, but summarize the whole genome in one number.
- **Allele dosage and segregation.** Gaynor et al. and Xie et al. use genotype classes (AAAB/ABBB versus AABB) or resequencing of many individuals to infer disomic versus polysomic inheritance directly.

ploidyspec sits between these. It uses chromosome-scale, haplotype-resolved assemblies of one individual, as produced at scale by the Darwin Tree of Life project, and reads structure chromosome by chromosome and along chromosomes, without annotation. It complements gene-based methods rather than replacing them.

3 What we are trying to achieve

3.1 The aim

Given a haplotype-resolved assembly, describe where the genome sits on two axes:

1. **Origin-like structure:** do the chromosome copies form one interchangeable pool (auto-like), or separate lineages at a consistent depth across the genome (allo-like), or something in between?
2. **Rediploidization:** has a once-tetrasomic genome started to split into lineages, where, by what mechanism (fusion, inversion, gradual divergence), and how far has it gone?

The tool should run on any species without tuning, report evidence rather than labels, and say when a question cannot be answered from the data.

3.2 Design principles

- **Reference-free and annotation-free.** Everything comes from k-mers of the assembled sequence, so a newly assembled species can be analysed the day its assembly is released.
- **Per chromosome, then per window.** Genome-wide summaries hide exactly the heterogeneity that rediploidization creates.
- **Descriptive states, not verdicts.** Each chromosome gets a state with the numbers that produced it.
- **Several independent lines of evidence.** Whole-chromosome distance, windowed distance, repeat markers, chromosome fusions and genome-wide partitions each fail differently, so agreement between them carries weight.
- **Validated on genomes with a known answer.** A simulator builds synthetic assemblies (diploid, auto, allo, rediploidizing, mislabelled) and an end-to-end test checks the whole pipeline against the truth.

4 The tool

4.1 Input

ploidyspec needs, for one individual, every haplotype at chromosome scale, described by a **manifest**: a two-column table of FASTA path and haplotype label.

```
# fasta_path      hap_label
data/sp/sp.hap1.1.primary.fa.gz  HAP1
data/sp/sp.hap2.1.primary.fa.gz  HAP2
```

Use `AUTO` instead of a label when one file holds several haplotypes named in the headers. Each chromosome-scale sequence (default ≥ 1 Mb) needs a chromosome number, read from the header: `chromosome: 4` in the description, or `SUPER_4` in the name. Other naming schemes are handled with `--chrom-regex/` `--hap-regex`. Shorter or unnumbered sequences are listed in `unplaced.tsv`, and the large ones are still tested as possible fused chromosomes.

What counts as 'a copy'

ploidyspec counts what the assembly contains. A tetraploid can be assembled as four haplotypes (four copies of each chromosome number) or as two haplotypes in which each base chromosome appears under two numbers (two copies each, arranged in homeolog pairs). Both are handled, and the answers to the four questions take the layout into account.

4.2 Installing and running

```
conda env create -f environment.yml && conda activate ploidyspec
pip install .
ploidyspec all --manifest manifests/sp.tsv --outdir results/sp --threads 8
ploidyspec all ... --with-te-markers --with-windowed-homeologs # optional stages
ploidyspec panel --results results --outdir meta # many species
```

The only external tools are `samtools` and `FastK` (`FastK`, `Logex`, `Histex`, `Tabex`, `Profex`). Every stage can be run on its own, and each skips work already done, so an interrupted run resumes. `report.html` in the output directory collects tables and plots from whatever has run.

Resources. Memory scales with the longest chromosome, not with genome size: about 40 MB per Mb for the matrix stage, and about 14 bytes per base of the longest chromosome per thread for the windowed stage. Runtime on our panel was 0.2-13 hours per species. k-mer tables dominate disk use; `--cleanup` removes them and keeps every result.

4.3 The measurement underneath everything

A **k-mer** is a word of k bases. ploidyspec builds a FastK table of every k-mer in each chromosome copy and compares tables. For two sequences A and B with Jaccard index $J = |A \cap B| / |A \cup B|$ of their k-mer sets, the Mash distance (Ondov et al., 2016) estimates the per-base divergence:

$$d = -\frac{1}{k} \ln \left(\frac{2J}{1+J} \right)$$

Short k-mers collide by chance and saturate, long ones are too sensitive. So the matrix stage computes d at several k (default 15 and 23) and keeps the largest k that clears the chance-collision floor. On all 53,309 chromosome pairs of our panel that was $k = 23$. For windows, ploidyspec uses **containment** c , the fraction of one sequence's k-mers present in the other, and the matching distance $d = -\ln(c)/k$.

Scale to keep in mind

Typical values in our panel: two alleles of the same chromosome, 0.0001 to 0.04; ancient homeologs, 0.05 to 0.10; unrelated chromosomes, 0.07 to 0.16. The confirmed diploid *Malva sylvestris* (ddMalSylv1) has alleles 0.0038 apart and unrelated chromosomes 0.116 apart, 31 times further.

5 The modules, one by one

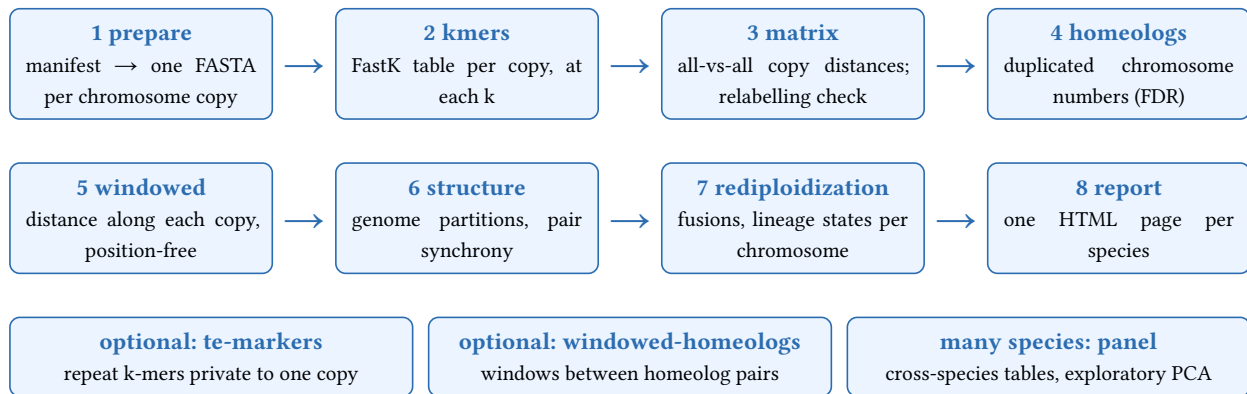


Figure 1: The pipeline. `ploidyspec all` runs stages 1-8 in order; each stage reads the outputs of the stages before it.

5.1 prepare

Reads the manifest, finds each chromosome-scale sequence, gives it a haplotype and a chromosome number, and writes `sequences.tsv` (one row per **unit**, a single chromosome copy such as `HAP2_chr05`) and `unplaced.tsv` (everything else, with the reason). If many sequences land in `unplaced.tsv`, the naming regex needs adjusting.

5.2 kmers

Extracts each unit to its own FASTA and builds a FastK table for each k. These are cached and reused by name. Each cached unit also records where it came from (assembly file, its size and modification time, sequence name and length), and is rebuilt if any of these change. This was added after a re-run on a new assembly silently reused the old sequences (Section 8.1).

5.3 matrix: copy distances and the relabelling check

Computes the distance between every pair of units, writes the full matrix, heatmaps, `ploidy_summary.tsv` (copies per chromosome number) and `homologous_chromosomes.tsv`.

It also checks **labelling**. Different haplotype assemblies are sometimes numbered differently: one file's "chr2" can be another file's "chr4". The check takes the file with the most units as reference. Each other file's units are matched one-to-one to the reference chromosome numbers, closest first. A unit is relabelled when its matched number is at least 3× closer than its declared one. A pair of units that each match the other's number (a swap) needs 2× on both sides. Ambiguous cases are listed in `matrix/chrom_label_corrections.tsv`, and relabelled units are renamed on disk so every later stage sees the corrected numbers.

5.4 homeologs: are some chromosome numbers duplicates of each other?

After a WGD, some chromosome numbers are each other's ancient copies (**homeologs**). For every pair of chromosome numbers this stage takes the mean distance over all copy combinations. It compares that mean to the background of all pairs using an empirical p-value, and controls the false discovery rate at 5%. A pair is accepted when it is significant and at least 2% closer than the background median. Outputs: `homeolog_pairs.tsv` (accepted pairs), `homeolog_candidates_ranked.tsv` (every pair, ranked) and `ploidy_ancestry_summary.tsv`. The last gives each chromosome's partner and `distance_ratio`, the homeolog distance divided by the allele distance.

5.5 windowed: what happens along each chromosome

Each copy is cut into 250 kb windows along its own coordinates. Each window is looked up, with FastK's profile mode, in each other copy's whole chromosome. That gives the fraction of the window's k-mer positions

found anywhere in the other copy, and the distance $-\ln(c)/k$ ($k = 15$). No alignment or shared coordinates are needed, so indels, inversions and gaps between assemblies do not matter. The result is one track per ordered pair of copies, along the first copy.

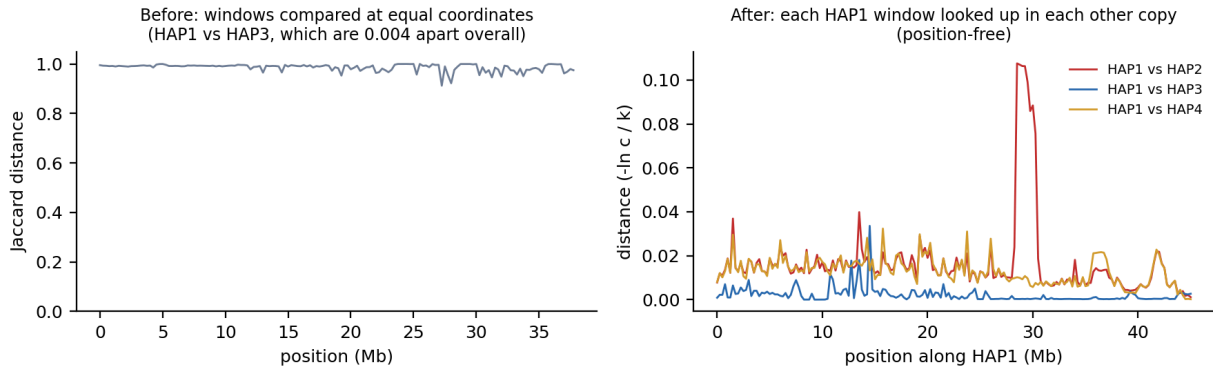


Figure 2: Why the windowed stage was rewritten. Left: the original method compared windows at the same coordinates in each copy. Once two assemblies differ by an indel larger than a window, every later window compares different sequence. Two copies of `ddEmpNigr1` chr03 that are 0.004 apart over the whole chromosome read as almost completely different (Jaccard ≈ 0.99) in almost every window. Right: the position-free method on the same chromosome. HAP3, the close copy, stays near zero all along, and HAP2 and HAP4 form the other lineage. The spike near 29 Mb is sequence present in HAP1 and absent from HAP2.

5.6 structure: genome-wide partitions and pair synchrony

`genome_partition.tsv` asks whether the chromosome numbers fall into groups that are closer within than between, more than chance, even when no single homeolog pair is significant. Chromosome numbers are clustered (average linkage) and, at every number of groups k , the between/within separation is compared with 500 random groupings of the same sizes. The randomness is seeded per species, so results reproduce. A strong, balanced two-group split (`daInuConz1`: 8 vs 8, $z = 10.6$) is a candidate subgenome structure that pairwise tests miss.

`pair_synchrony.tsv` gives the spread of divergence depth across the accepted homeolog pairs (`pair_depth_cv`). One duplication event diverges every pair to about the same depth (low CV, `daGleHede1` 0.04). Several events, or pairs that stopped recombining at different times, spread them out.

5.7 te-markers (optional): fossil repeat activity

See Section 6.3 for the idea. For each pair of copies of a chromosome, at $k = 13$, the stage finds **high-copy** k-mers (count ≥ 100 , mostly transposable elements and other repeats). It then finds **markers**: high-copy k-mers at least $2\times$ more abundant in one copy than the other. The summary is

$$\text{te_marker_fraction} = \frac{m_A + m_B}{h_A + h_B}$$

with m the markers and h the high-copy k-mers of copies A and B. `te_marker_fraction_by_lineage.tsv` splits chromosomes with three or more copies into two groups by distance and reports `split_ratio`: the cross-group fraction over the within-group fraction. Copies with too little or too much repeat content for their length are flagged as assembly problems. The method follows K. Jaron's fossil-TE approach for separating subgenomes.

5.8 rediploidization: fusions and per-chromosome states

The stage that combines everything. It has three parts.

Fusion detection. It tests two kinds of sequence:

- placed copies at least $1.4\times$ the median length of their sibling copies;
- unplaced scaffolds at least half the median chromosome length.

For each, it measures how much of every other chromosome's k-mer content it contains. A chromosome counts as a fused component when its containment is far above background (robust $z \geq 10$, $\geq 2\times$ the background median, and $\geq$ a quarter of the top hit). For placed copies, the partner must also be $\geq 2.5\times$ more enriched, per k-mer, than in the copy's unfused siblings (`sibling_excess`). This rules out ancient homeology, which every copy shares, and siblings that are merely fragments. The scaffold is called a `fusion` only if the partner chromosome is missing from that haplotype. `lineage_divergence` $= -\ln(c)/k$ dates the fused lineage against the unfused copies.

Lineage structure per chromosome. For each chromosome number with three or more copies it reads:

- `dist_split`: split the copies into the two most separated groups; mean cross-group over mean within-group distance;
- `te_split`: the TE-marker `split_ratio` for the same chromosome;
- `split_extent`: along each copy, which windows show the split. Reported as `whole`, `regional` (with the segments) or `none`, from the median copy. It counts only if it beats a null: the same scan for groupings of the copies that cut across the observed one, pooled over the genome (Section 6.4).

Pooling. A two-haplotype assembly has only two copies of each chromosome number, too few to split. ploidyspec pools a chromosome with its accepted homeolog partner (four copies). Failing that, it uses its reciprocal best match across a significant two-group genome partition, and reads the pooled group.

Each chromosome then gets one `copy_state`:

State	Meaning
<code>fusion_lineages</code>	Fused in some haplotypes and not others: the fused and unfused copies form separate lineages (the snow carp mechanism).
<code>resolved_lineages</code>	Balanced split, whole-chromosome split $\geq 2.5\times$, holding along most of the chromosome; or, for a pooled group, the two numbers are separate lineages.
<code>partially_resolved</code>	Two or more of: distance split ≥ 1.25 , TE split ≥ 2 , a windowed split segment.
<code>candidate</code>	Exactly one of those.
<code>tetrasomic_like</code>	None: the copies are interchangeable. For a pooled group, the homeolog is as close as the allele.
<code>one_divergent_copy</code>	One copy stands apart from the rest along the chromosome (distance or TE split; a windowed segment alone does not count). This is an odd haplotype (assembly quality, or a divergent extra genome, AAAB-like), not two lineages. Every split of a triploid looks like this.
<code>not_assessable</code>	Fewer than three copies and nothing to pool with.

5.9 report, panel, simulate, cleanup

`report` writes one self-contained HTML page per species, with core sections first and TE markers under "Supplementary". `panel` stacks every species into `panel_summary.tsv`, `genome_partition.tsv` and `rediploidization_panel.tsv`. It also writes a TE table and an exploratory PCA over the core numbers in `supplementary/`. `simulate` writes synthetic assemblies with a known answer, and `cleanup` deletes the large intermediate files.

6 Answering the four questions

This chapter is the practical guide: given a finished run, how to read the outputs to answer each question, and how much to trust the answer.

6.1 Question 1: what is the ploidy?

Assembly layout	Copies per number	Homeolog pairs	Reading
Two haplotypes, no pairs	2	none	Diploid-like. (Or a polyploid whose duplicate sets have diverged beyond detection; check <code>genome_partition.tsv</code> .)
Two haplotypes, numbers in pairs	2	most numbers	Tetraploid assembled as two haplotypes: 2 copies $\times$ 2 sets. <code>daGleHede1</code> : 18 numbers in 9 pairs.
Four haplotypes, no pairs	4	none	Tetraploid, all four copies in view.
Four haplotypes, numbers in pairs	4	most numbers	Octoploid on half the numbers: <code>ddLepDrab1</code> , $4 \times 16 = 64$ chromosomes, 8 pairs, $2n = 8x$ on $x = 8$.
Three haplotypes	3	-	Triploid (<code>ddSalTrial</code>).

Table 1: Reading ploidy from copies and homeolog pairs.

Base number x is the number of chromosome numbers divided by the sets per haplotype, and ploidy is copies $\times$ sets. **Confidence is high for the copy count**, which is set by the assembly. That is also its limit: a collapsed or missing haplotype lowers the count. `ddAraThal4` has only one usable haplotype, so its copy count says nothing about ploidy. **Confidence in the number of sets** depends on how many chromosome numbers fall into accepted pairs, and on whether `ploidy_summary.tsv` is uniform. Check counts against the literature (`meta/literature_ploidy.tsv`) when possible.

Mislabelled chromosomes make ploidy look strange

If one haplotype file numbers chromosomes differently, every comparison of “chr2” mixes two chromosomes. Always check `matrix/chrom_label_corrections.tsv`. Corrected rows are fixed automatically; an `ambiguous` row is a warning to look by hand.

6.2 Question 2: auto-like or allo-like, and with what confidence?

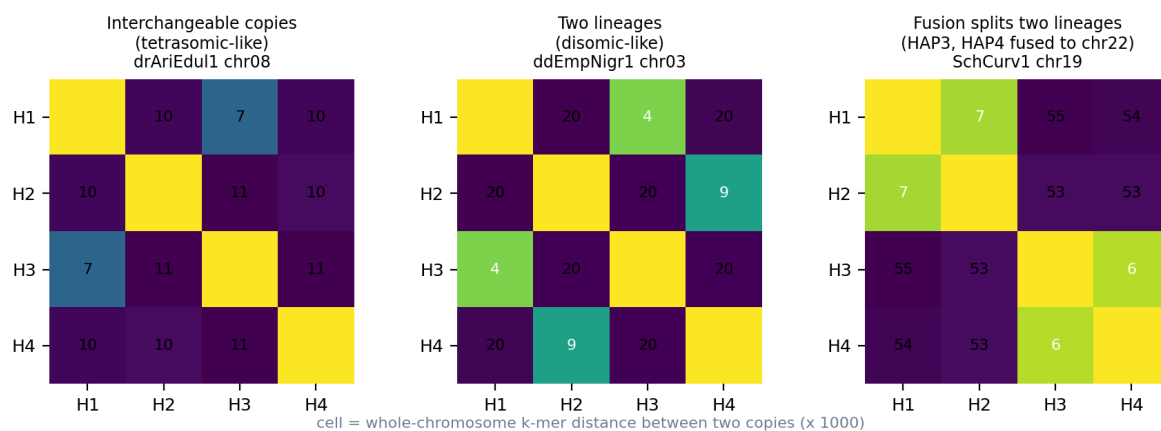


Figure 3: The three shapes that matter, from real chromosomes (whole-chromosome distance between each pair of copies, $\times$ 1000). Left: all six distances similar, so the copies are one interchangeable pool. Middle: two tight pairs at a uniform distance from each other, so two lineages. Right: two lineages whose split coincides with a fusion carried by two of the four copies.

Read the evidence in this order.

1. **Within each chromosome** (three or more copies): are the copies interchangeable or split into two lineages? This is `copy_state`, with `dist_split` behind it.
2. **Across chromosomes**: is the split at the same depth on every chromosome? An allo-like genome splits everywhere, at a consistent depth (one origin). A rediploidizing genome mixes tetrasomic-like and split chromosomes, at different depths.
3. **Between chromosome numbers** (two-haplotype assemblies): accepted homeolog pairs and their `pair_depth_cv`; strong balanced genome partitions.
4. **Repeat history**: a TE-marker split that agrees with the distance split (`te_split` ≥ 2 on the same chromosomes).

Reading	Evidence pattern	Confidence
Allo-like (disomic-like) throughout	Most chromosomes <code>resolved_lineages</code> , at similar depth; or (two haplotypes) most numbers in homeolog pairs with low <code>pair_depth_cv</code> ($\approx 0.03-0.05$), resolved when pooled; TE split agrees.	High when two or more lines agree across most chromosomes.
Auto-like (tetrasomic-like) throughout	Most chromosomes <code>tetrasomic_like</code> : all copies about equally close; no lineage split by distance, TE or windows.	High with four or more copies. Not available from two haplotypes alone.
Mixed / transitional	Some chromosomes tetrasomic-like, others split or partially split; fusions; splits at different depths.	Medium : the per-chromosome pattern is solid, the mechanism needs checking.
Cryptic subgenome structure	No accepted pairs, but a strong balanced genome partition ($z \geq 10$).	Medium : one line of evidence; confirm with karyology.
Not assessable	Two haplotypes, no pairs, no partition; or one usable haplotype.	-

Table 2: How to read origin-like structure, with confidence. “Lines of evidence” are whole-chromosome distance, windowed extent, TE markers, homeolog pairs and genome partitions.

Four things that look allo-like but are not

- **A rediploidized autopolyploid**. It splits into lineages exactly like an allopolyploid: the snow carp fused chromosomes; Twyford et al. Box 2.
- **One odd haplotype**. A copy assembled less completely than its siblings, or a mislabelled haplotype, isolates one copy (`one_divergent_copy`). If the same haplotype is odd on most chromosomes, suspect the assembly (`most_frequent_outlier_hap` in the summary).
- **Pooled resolved lineages**. Any two different chromosome numbers are far more distant than two alleles. So pooling a chromosome with its partner reads `resolved_lineages` even in a diploid (we checked on `ddMalSylv1`: splits of 22-47 $\times$). For pooled groups the evidence is the homeolog test or the partition z , and the informative outcome is the opposite one, `tetrasomic_like`.
- **Absolute `te_marker_fraction`**. See Section 6.3.

6.3 Question 3: TE markers, explained

The idea. Transposable elements (TEs) copy themselves around the genome in bursts. Two lineages that evolve apart, such as the parents of an allopolyploid, have separate bursts. Each lineage ends up with its own recently amplified repeats, and those stay behind as a “fossil” signature. In an allopolyploid, a subgenome carries its parent’s repeats and the other subgenome largely does not. So many abundant repeat k-mers are much more common in one copy than in the other: high `te_marker_fraction`. In an autopolyploid, all copies come from one genome with one repeat history, so few should be.

What we found. The confirmed diploid `ddMalSylv1` has a per-chromosome `te_marker_fraction` of 0.05-0.25 (median 0.14). Ordinary allelic differences produce markers too: TE insertions present in one allele but not the other. This range overlaps the allopolyploid anchors (`daGleHede1` median 0.22, `drTriRepe1` 0.15). The autotetraploid snow carp `SchCurv1` spans 0.07 to 0.61.

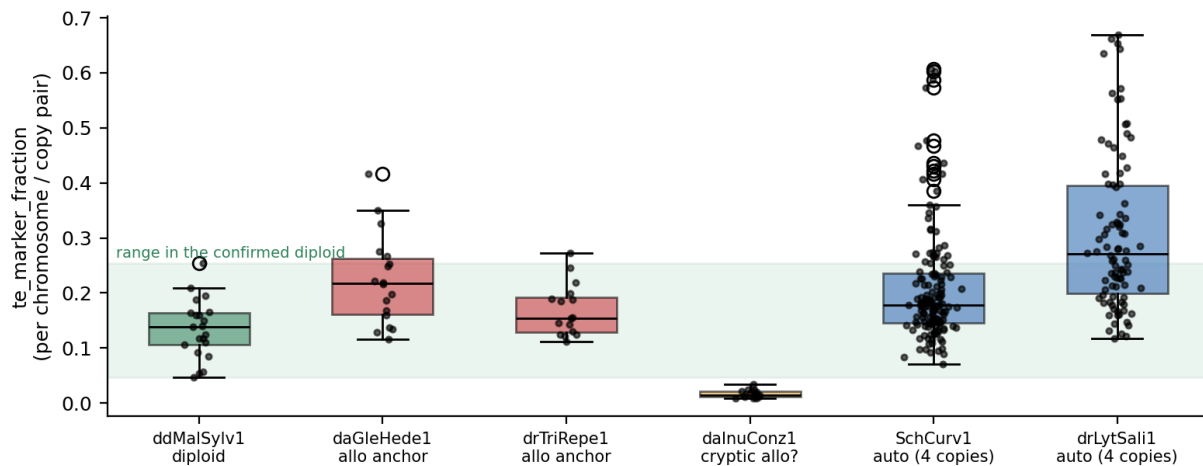


Figure 4: Absolute `te_marker_fraction` does not separate diploid, auto- and allopolyploid. Each point is one chromosome copy pair. The shaded band is the range in the confirmed diploid. `daInuConz1` is low because it compares allelic copies within each subgenome; its subgenome split lies between chromosome numbers, which this index never compares.

How to use TE markers. Use them as a **contrast within one genome**. For a chromosome with three or more copies, split the copies by distance and ask whether markers are concentrated between the two groups (`split_ratio` ≥ 2) rather than within them. When the TE split and the distance split agree, the lineages have separate repeat histories, an independent line of evidence. `SchCurv1` chr19, the fused chromosome, has a TE split of 5.6 against 0.8-2.2 for most of its genome.

Reading `te_split` in practice

- `te_split` ≈ 1 : repeats do not track the lineages.
- `te_split` ≥ 2 on the same chromosomes as `dist_split` ≥ 1.25 : two lineages with separate repeat histories.
- High absolute fractions with `te_split` ≈ 1 : allelic TE polymorphism, or an assembly artefact. Check `flagged_units`.

6.4 Question 4: is there evidence of rediploidization?

Rediploidization leaves four kinds of trace that `ploidyspec` can see:

1. **Chromosome fusions between copies:** `fusions.tsv` rows with `status = .` These are the strongest fusion single evidence: a copy that has absorbed another chromosome cannot pair freely with its unfused siblings. Their `lineage_divergence` orders the fusions in time.
2. **Chromosomes whose copies have split into lineages** in a genome where others have not: a mixture of `resolved_lineages` / `fusion_lineages` and `tetrasomic_like`.
3. **Splits along part of a chromosome:** `partially_resolved` with a `regional` extent. This is the pattern of a centromeric inversion (snow carp chr17), or of divergence spreading out from a fusion site.
4. **Ancient pairs at different depths:** a high `pair_depth_cv`. Duplicate pairs that stopped recombining at different times, such as `drMyrSpic1` at 0.33 against 0.03-0.05 for single-event anchors.

Reading	Pattern
Rediploidization under way	Fusions and/or a mix of split and tetrasomic-like chromosomes in a genome with three or more copies.
Completed (or allo-like)	Every chromosome split at a consistent depth; pairs synchronous.
Not started	All chromosomes tetrasomic-like, no fusions.
Partial or ongoing on a few chromosomes	Regional splits on a minority of chromosomes. Check that segments are not noise (see the caution below).

Table 3: Reading rediploidization.

How regional splits are kept honest

Window distances are real measurements with real noise, so a run of split windows can occur by chance. Two rules guard against it.

- **A null for each species.** The window scan is repeated for groupings of the copies that cut across the observed one: for a balanced split, the other balanced splits; for a one-copy split, the splits isolating each other copy. Their coverage, pooled over the genome, gives the species' noise level. A split's extent counts only if it covers more of the chromosome than that null's 95th percentile and more than every crossing grouping of the same chromosome (`window_covered`, `window_null_q95`, `split_extent_raw` show the working). In `daBudDavi1` this removed ten calls resting on single 2-3 Mb segments. Snow carp chr19 (0.89 against a null of 0.07) and chr17 (0.24) pass, as does the simulated regional split.
- **One copy apart needs chromosome-wide evidence.** In a polysomic genome, one of four copies often carries a divergent haplotype block somewhere. That is ordinary polymorphism, not a lineage. So `one_divergent_copy` requires a distance or TE split across the chromosome. On regional segments alone, `drLytSalil`, tetrasomic by classical genetics, read 12/15 chromosomes as `one_divergent_copy`; it now reads 13/15 `tetrasomic_like`.

7 What we have found so far

7.1 Validation on simulated genomes

`ploidyspec simulate` builds synthetic assemblies with a known answer: random sequence with TE families inserted at high copy number. Every haplotype except HAP1 carries a transposition and an inversion, so the copies are not collinear, as with real independently assembled haplotypes. The end-to-end test runs the whole pipeline on each and checks the result.

Scenario	What it contains	Chromosomes correct
diploid	two haplotypes of one genome	7 / 7
autotetraploid	four equivalent haplotypes	7 / 7
autotetraploid, 2 haplotypes	the same, assembled as two haplotypes with each chromosome numbered twice	14 / 14
allotetraploid	two diverged progenitors, each with its own TE burst	14 / 14
rediploidized	autotetraploid with a whole-chromosome lineage split, a split over 40% of one chromosome, one placed and one unplaced fusion	7 / 7, both fusions found
mislabelled	autotetraploid with one haplotype numbered shifted by one, and a swap in another	7 / 7, all 9 relabelled

Table 4: End-to-end validation (11 tests, all passing). On the rediploidized scenario, the pre-fix windowed code calls chr07 `one_divergent_copy` and loses chr02's regional split, so the test now guards against that bug.

7.2 The snow carps: blind recovery of a published result

- **Fusions.** In `SchYoun1` all five published fusions are recovered from unplaced scaffolds. In `SchCurv1`, chr19+chr22 is fused in HAP3 and HAP4 (containment 0.44-0.45, $z \approx 48$, $3.4\times$ enriched over unfused siblings).
- **Their order in time.** `lineage_divergence` ranks the `SchYoun1` fusions as chr19+22 (≈ 0.053), then chr04+15, chr08+16 and chr20+23 (≈ 0.026 -0.028), then chr11+14 (≈ 0.015). This is the three-wave order Xie et al. inferred, found without knowing it.
- **Lineage split.** The fused chromosome splits along its whole length, in 92% of windows.
- **chr17.** The analysis flagged chr17 unprompted. With the corrected windows, every copy places the split in the last ≈ 8 Mb of the chromosome. Xie et al. report a short-arm-only disomic shift caused by a centromeric inversion; whether that 8 Mb block is the short arm still has to be checked against their coordinates.

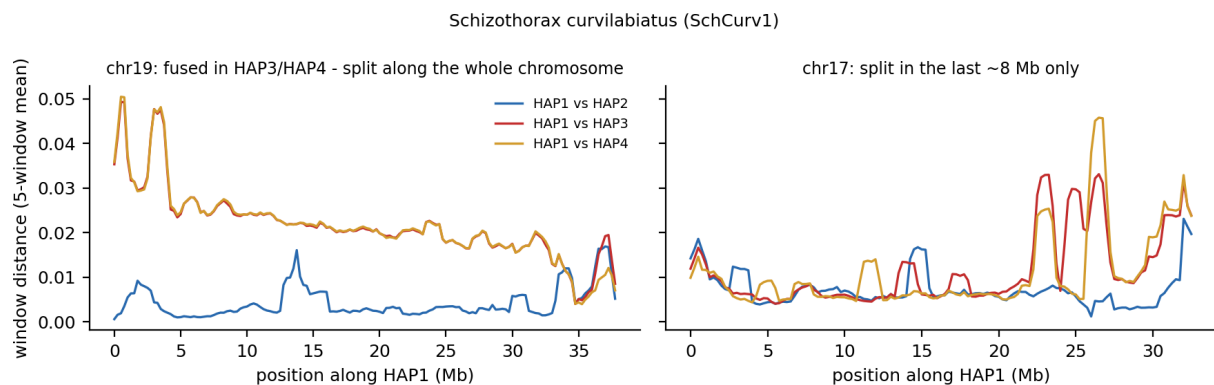


Figure 5: Window distances along HAP1 of `SchCurv1`. chr19: the fused copies (HAP3, HAP4) are further from HAP1 than HAP2 is, along the whole chromosome. chr17: the copies separate only towards the end.

7.3 The panel: 45 assemblies

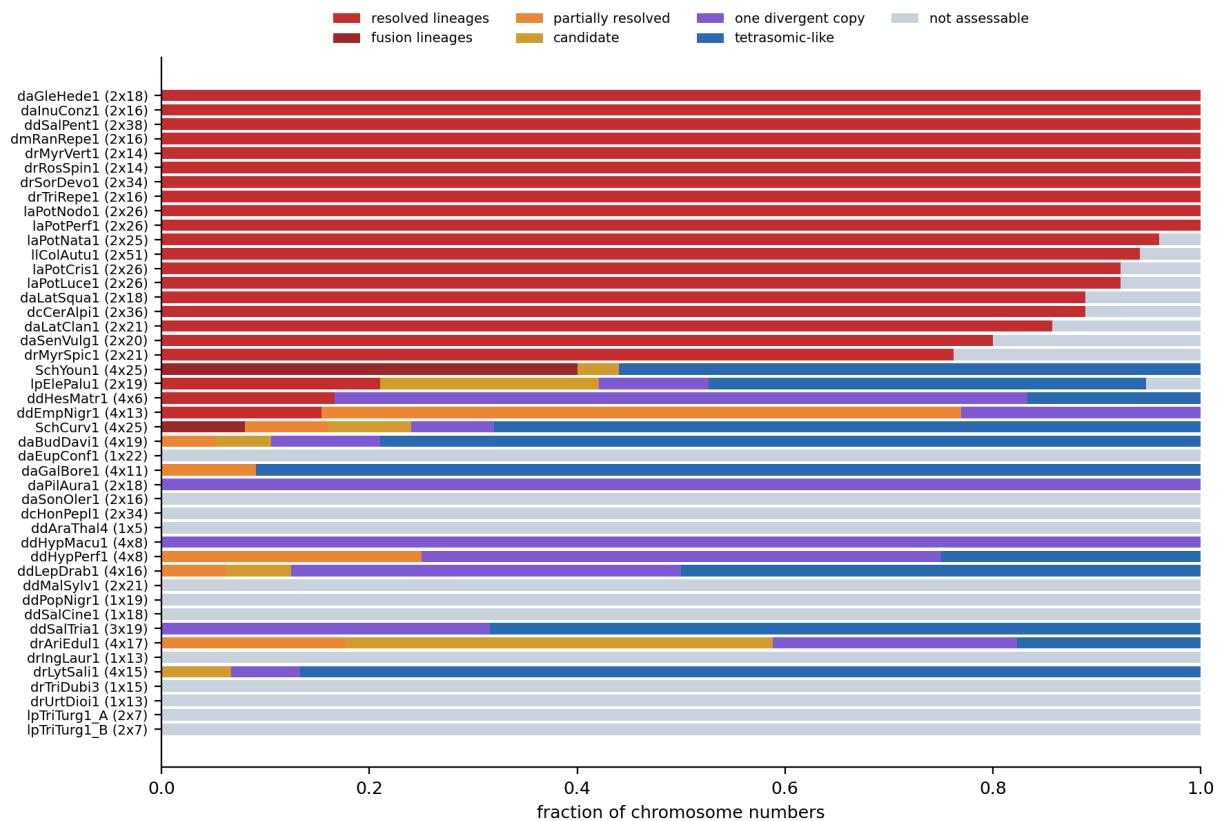


Figure 6: Rediploidization state of each chromosome number, per species (label: copies × chromosome numbers), after the corrections in Section 8.1. `daInuConz1`, `dmRanRepe1` and `dcCerAlpi1` read `resolved_lineages` through partition pooling: the evidence there is the partition `z`, not the bar.

Two-haplotype allopolyploid anchors (`daGleHede1`, `drTriRepe1`, `drSorDevo1`, `ddSalPent1`, `drMyrVert1`, `drRosSpin1`, the *Potamogeton* species) have most chromosome numbers in homeolog pairs, synchronous pair depths (`pair_depth_cv` 0.03-0.05 for the clearest), strong genome partitions, and resolved pooled lineages. That is what a single, old, disomic duplication looks like.

Cryptic subgenome structure. `daInuConz1` (*Pentanema conyzae*) has no accepted homeolog pair but an 8 vs 8 partition at $z = 10.6$. That matches its documented tetraploid count $2n = 4x = 32$ on $x = 8$. `dmRanRepe1` (8 vs 8, $z = 10.9$) and `dcCerAlpi1` (16 vs 20, $z = 18.9$) show the same kind of signal.

Autopolyploid anchors read as they should. `drLytSali1`, tetrasomic by classical genetics, is 13/15 `tetrasomic_like` on its current assembly. The **snow carps** combine fusions and whole-chromosome lineage splits with a tetrasomic-like majority (`SchCurv1` 17/25, `SchYoun1` 14/25 plus 10 fusion-lineage chromosomes): rediploidization in progress.

Two lineages per chromosome. `ddEmpNigr1` shows, on most chromosomes, two lineages at a uniform cross distance (≈ 0.020). That suggests a single origin with disomic-like structure; it is now 2 `resolved` and 8 `partially_resolved` of 13.

One divergent copy. `daPilAural` shows one copy in four apart on all 9 base chromosomes, with the odd copy changing haplotype. That is AAAB-like. `ddHypMacu1` has the same haplotype (HAP1) apart on all 8 chromosomes, matching a known assembly problem in that haplotype.

An octoploid, read correctly at last. `ddLepDrab1` has four haplotypes of 16 chromosome numbers in 8 homeolog pairs at a uniform depth (CV 0.06). Within each number the four copies are mostly tetrasomic-like. That fits two $x = 8$ sets each present four times (AAAABBBB-like), matching $2n = 64$ on $x = 8$. Its earlier “strongest allopolyploid candidate” signals came from one haplotype file numbering 14 chromosomes differently (Section 8.1).

Species	Copies × nos.	Allele dist. (median)	Pairs	Pair depth CV	Partition z (k)	Reading
daGleHede1	2 × 18	0.014	9	0.042	23.8 (9)	allo anchor; one synchronous event
drSorDevo1	2 × 34	0.011	17	0.030	42.5 (17)	allo anchor
ddSalPent1	2 × 38	0.009	19	0.038	47.2 (19)	allo anchor
drMyrSpic1	2 × 21	0.009	8	0.332	24.7 (7)	hexaploid; asynchronous pairs
daInuConz1	2 × 16	0.0007	0	-	10.6 (2)	cryptic 8 vs 8 split, x = 8
dmRanRepe1	2 × 16	0.023	0	-	10.9 (2)	cryptic 8 vs 8 split
dcCerAlpi1	2 × 36	0.008	0	-	20.8 (3)	cryptic 16 vs 20 split
ddMalSylv1	2 × 21	0.0038	0	-	none	diploid anchor
drLytSali1	4 × 15	0.025	5	0.047	10.5 (5)	auto anchor; 13/15 tetrasomic-like
SchCurv1	4 × 25	0.013	2	-	-	auto; 1 fusion; 17/25 tetrasomic-like
SchYoun1	4 × 25	0.017	0	-	-	auto; 5 fusions; 14/25 tetrasomic-like
ddLepDrab1	4 × 16	0.021	8	0.062	18.8 (8)	octoploid, AAAABBBB-like on x = 8
ddEmpNigr1	4 × 13	0.020	1	-	-	two lineages per chromosome
daPilAur1	2 × 18	0.039	9	0.105	22.7 (9)	one divergent copy in four

Table 5: Selected species from `panel_summary.tsv`. Pairs: accepted homeolog pairs. Partition: strongest genome partition. Allele distance: median distance between copies of the same chromosome number.

8 Caveats, corrections and open questions

8.1 What we got wrong, and fixed

Checking results by hand during Phase 2 found four problems. All four are now fixed and covered by tests. They are listed here because earlier write-ups quoted numbers that depended on them.

Problem	What happened	Fix
Window registration	Windows were compared at equal coordinates. After the first indel larger than a window they compared different sequence, so across the panel only 0-34% of windows were usable. Every windowed-derived number was affected: regional splits, <code>split_extent</code> , run-length and windowed-CV metrics.	Position-free windows (FastK profiles). Panel re-run.
Chromosome relabelling	<code>ddLepDrab1</code> HAP4 numbers 14 of 16 chromosomes differently. The check rejected the whole batch because two units tied, so HAP4 stayed mislabelled. Its “strongest allo candidate” signals were all artefacts. <code>ddHesMatr1</code> has a HAP1 chr01/chr02 swap below the old threshold; its only homeolog pair was the swap.	One-to-one matching; mutual swaps at 2×. Both re-run.
Stale cache	Per-copy FASTAs and k-mer tables were reused by name. A re-run of <code>drLytSalil</code> on a new, renumbered assembly kept the old sequences.	Cache records its source; rebuilt on any change.
Window noise	After the windowed fix, chance runs of split windows and ordinary haplotype blocks were read as lineage splits, inflating <code>one_divergent_copy</code> and <code>candidate</code> .	Null from crossing groupings; one copy apart needs chromosome-wide evidence (Section 6.4).

Several inheritance-mode metrics from earlier work (`partition_consistency`, run length, `flip_rate`, the windowed CVs and a combined score) were retired, because labelling and window registration drove them. Their last values are archived in `meta/archive/` with a note.

8.2 Limits of the approach

- **One individual.** Inheritance and origin are not observed, only structure.
- **Assembly quality is part of the measurement.** Missing or extra sequence in one haplotype reads as divergence: a copy missing 10% of a chromosome reads as the divergent copy, with a regional split. Compare lengths and `unplaced.tsv` before reading biology into one odd haplotype.
- **Old duplications fade.** Homeolog distances approach the background of unrelated chromosomes, and k-mer methods lose them. Ancient WGDs such as the land snail event (McHale et al., 2025) are better seen with gene-based approaches (duplicated BUSCOs, synteny, Ks).
- **Stay-tetrasomic autopolyploids leave little divergence** (Gaynor et al., 2026). They are visible only when the assembly shows all copies.
- **Homoeologous exchange** can make allopolyploid chromosomes look locally auto-like (Deb et al., 2023); regional splits in an allopolyploid deserve that reading.
- **Thresholds are provisional**, tuned on the snow carps and on simulations. They are all exposed as command-line options.

8.3 Where ploidyspec sits in poly-space

Twyford et al. (2025) propose placing polyploids in a multidimensional space of measurable genomic features. ploidyspec provides several axes from one assembly:

Twyford et al. axis	ploidyspec measurement
Heterozygosity / segregating variation	<code>allelic_distance_median</code> : distance between copies of the same chromosome number

Twyford et al. axis	ploidyspec measurement
Sequence divergence between duplicated sets	homeolog pair distance, <code>distance_ratio</code> , <code>dist_split</code>
Structural divergence	fusions between copies; regional lineage splits
Temporal heterogeneity (diploidization)	mix of states across chromosomes; <code>pair_depth_cv</code> ; fusion <code>lineage_divergence</code>
TE distribution	<code>te_split</code> (within-genome contrast)

`ploidyspec panel` builds an exploratory PCA over these numbers (`meta/supplementary/poly_space_pca.png`). It is a picture, not a classifier.

8.4 Open questions

- Confirm the `SchCurv1` chr17 block against the published coordinates.
- Literature checks for the cryptic candidates (`dmRanRepe1`, `dcCerAlpi1`).
- Assess two-haplotype genomes for tetrasomy more directly: the pooled test can show tetrasomy, but cannot show allopolyploidy.
- Add anchors with mapped rediploidization, ideally a salmonid.
- Auto-detect chromosome and haplotype naming, so manifests need no regexes.

9 Output reference

File	What it holds
<code>sequences.tsv</code> , <code>unplaced.tsv</code>	one row per chromosome copy (unit); everything else, with the reason
<code>matrix/whole_chrom_pairs.tsv</code>	distance (and k-mer counts) for every pair of units
<code>matrix/ploidy_summary.tsv</code>	copies per chromosome number and their haplotypes
<code>matrix/chrom_label_corrections.tsv</code>	relabelled and ambiguous units
<code>matrix/*heatmap*.png</code>	the full distance matrix
<code>homeologs/homeolog_pairs.tsv</code>	accepted homeolog pairs (FDR 5%, $\geq 2\%$ below background)
<code>homeologs/homeolog_candidates_ranked.tsv</code>	every pair of chromosome numbers, ranked, with z, p, q
<code>homeologs/ploidy_ancestry_summary.tsv</code>	each chromosome's partner and <code>distance_ratio</code>
<code>homeologs/homeolog_pairs_reordered_heatmap.png</code>	chromosome numbers clustered; pairs outlined
<code>windowed/windowed_all.tsv</code>	per window of <code>unit_a</code> against <code>unit_b</code> : <code>kmers_a</code> , <code>shared</code> , <code>containment</code> , <code>distance</code>
<code>structure/genome_partition.tsv</code>	significant k-group partitions of chromosome numbers, with z
<code>structure/pair_synchrony.tsv</code>	<code>n_accepted_pairs</code> , <code>pair_depth_median</code> , <code>pair_depth_cv</code>
<code>subgenomes/auto_allo_index.tsv</code>	<code>te_marker_fraction</code> per pair of copies
<code>subgenomes/te_marker_fraction_by_lineage.tsv</code>	within/cross-lineage fractions and <code>split_ratio</code>
<code>rediploidization/fusions.tsv</code>	tested scaffolds: components, containment, <code>sibling_excess</code> , <code>lineage_divergence</code> , <code>status</code>
<code>rediploidization/rediploidization_by_chrom.tsv</code>	per chromosome: splits, extent, segments, <code>copy_state</code> , <code>state_basis</code> , partner
<code>rediploidization/rediploidization_summary.tsv</code>	counts per state, fusions, outlier haplotype, pair depths
<code>report.html</code>	everything above, one page
<code>meta/panel_summary.tsv</code> (panel)	one row per species: the core numbers

Full column-by-column documentation with worked examples is in `OUTPUTS.md` in the repository.

10 Glossary

Allele distance k-mer distance between copies of the same chromosome number; a measure of heterozygosity.

Containment fraction of one sequence's k-mers found in another.

Copy / unit one assembled copy of one chromosome (HAP2_chr05).

Disomic inheritance each chromosome pairs with a preferred partner; copies form fixed pairs.

Fossil TE marker a repeat k-mer much more abundant in one copy than another, left by separate bursts of TE activity.

Fusion two chromosomes joined into one in some haplotypes.

Homeolog a chromosome derived from the same ancestral chromosome through polyploidy, in a different set.

Homoeologous exchange recombination between homeologs, swapping or replacing segments.

Lineage split copies of one chromosome falling into two groups that are closer within than between.

Mash distance per-base divergence estimated from shared k-mers.

Ohnolog a gene duplicated by whole-genome duplication.

Pooling reading a chromosome together with its homeolog partner when each has too few copies on its own.

Rediploidization return of a polyploid genome to diploid-like (disomic) behaviour, chromosome by chromosome.

Tetrasomic inheritance all four copies pair at random; copies stay interchangeable.

11 References

- Becher, H. et al. (2020). Tetmer: k-mer spectrum model for tetraploid genomes. As cited in Twyford et al. (2025).
- Deb, S. K., Edger, P. P., Pires, J. C. & McKain, M. R. (2023). Patterns, mechanisms, and consequences of homoeologous exchange in allopolyploid angiosperms: a genomic and epigenomic perspective. *New Phytologist* 238(6): 2284-2304. doi:10.1111/nph.18927
- Gaynor, M. L., Feng, K., Soltis, D. E., Soltis, P. S. & Smith, S. A. (2026). All detectable ancient whole-genome duplications involve hybridization. *bioRxiv* preprint, posted 28 August 2026, not peer reviewed. doi:10.64898/2026.08.25.747000
- Jaron, K. S. Separate sub-genomes of an allopolyploid. k-mer approaches for biodiversity genomics (wiki). github.com/KamilSJaron/k-mer-approaches-for-biodiversity-genomics
- Li, Z. et al. (2021). Patterns and processes of diploidization in land plants. *Annual Review of Plant Biology* 72: 387-410.
- Lv, Z., Addo Nyarko, C., Ramtekey, V., Behn, H. & Mason, A. S. (2024). Defining autopolyploidy: cytology, genetics, and taxonomy. *American Journal of Botany* 111(8): e16292.
- Mason, A. S. & Wendel, J. F. (2020). Homoeologous exchanges, segmental allopolyploidy, and polyploid genome evolution. *Frontiers in Genetics* 11: 1014.
- McHale, F., Mulhair, P. O. & Holland, P. W. H. (2025). Evolution of duplicated Hox gene clusters in land snails and slugs. *Journal of Experimental Zoology Part B* 344(6): 363-368. doi:10.1002/jez.b.23322
- Myers, G. FastK: a k-mer counter for high-fidelity shotgun datasets. github.com/thegenemyers/FASTK
- Ondov, B. D. et al. (2016). Mash: fast genome and metagenome distance estimation using MinHash. *Genome Biology* 17: 132.
- Ranallo-Benavidez, T. R., Jaron, K. S. & Schatz, M. C. (2020). GenomeScope 2.0 and Smudgeplot for reference-free profiling of polyploid genomes. *Nature Communications* 11: 1432.
- Twyford, A. D., Conover, J. L., Doyle, J. J., Mason, A. S., Soltis, D. E., Soltis, P. S. & Wendel, J. F. (2025). The polyploid continuum and the landscape of polyploid genomic variation. *American Journal of Botany* 112(11): e70121. doi:10.1002/ajb2.70121
- Wang, J., Qin, J., Sun, P., et al. (2019). Polyploidy index and its implications for the evolution of polyploids. *Frontiers in Genetics* 10: 807. doi:10.3389/fgene.2019.00807
- Xie, C., Ma, Z., Zhou, C., Ma, K., et al. (2026). Chromosomal fusions trigger rediploidization of autopolyploid genomes. *Nature* 654: 706-713. doi:10.1038/s41586-026-10439-1